

Chapter 1

Introduction

Diabetes is a very bad health problem. It happens for people all over the world, all genders, all ages. The notable problem in diabetes is the high level of blood glucose level.

Hospital readmissions among diabetic patients pose a significant burden on healthcare systems. Readmissions within 30 days of discharge are particularly concerning as they may indicate inadequate initial treatment, premature discharge, or insufficient follow-up care. Understanding the factors associated with readmission patterns is very important for developing effective intervention strategies to improve patient health and reduce healthcare costs.

This analysis utilizes a comprehensive dataset containing records of diabetic patient encounters at more than hundred hospitals in the US. The dataset comprises 101,766 observations across 50 variables, encompassing patient demographics, admission details, diagnoses, laboratory tests, and medication regimens. The target variable, "readmitted," categorizes patient outcomes into three classes: readmitted within 30 days (" <30 "), readmitted after 30 days (" >30 "), or not readmitted ("NO").

The primary objectives of this analysis is to identify patterns and relationships within the diabetic patient data through comprehensive exploratory data analysis (EDA). The most important part of this case study 2 is the methodology to handle missing values and impute them in the dataset. Subsequently, develop a predictive model using logistic regression with k-fold cross-validation to estimate readmission probabilities then evaluate the performance of the model and identify the most influential features affecting readmission outcomes. Lastly, determine the key factors associated with hospital readmission among diabetic patients.

Through this investigation, we aim to contribute insights that may inform clinical decision-making and intervention strategies to reduce readmission rates among diabetic patients.

Chapter 2

Data

The exploratory data analysis began with an examination of the dataset structure and composition. The diabetic dataset comprises 101,766 patient encounters with 50 variables.

The distribution of the target variable "readmitted" reveals that 53.9% patients were not readmitted, 34.9% were readmitted after 30 days, and 11.2% were readmitted within 30 days (Figure 2.1). The target variable is imbalanced, which is a common issue in healthcare datasets.

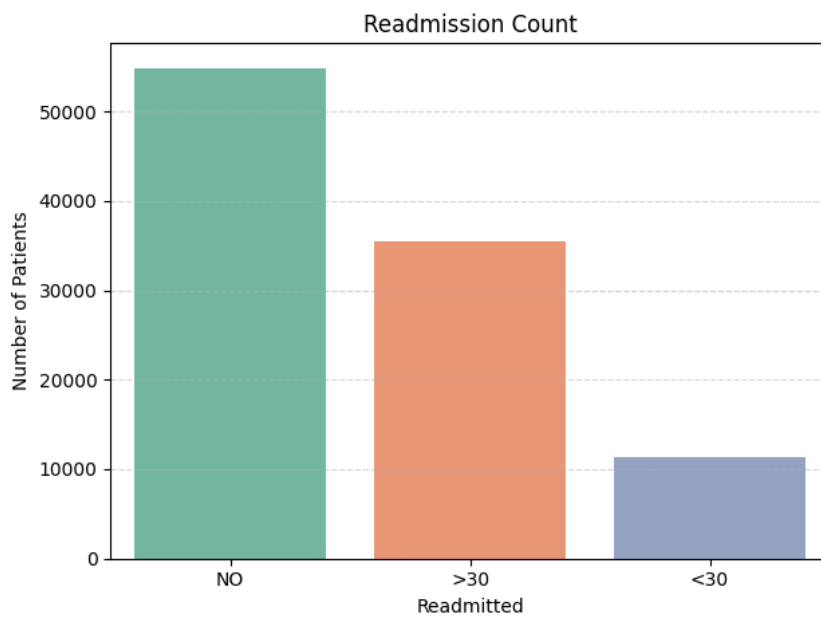


Figure 2.1: Distribution of patient readmission categories

A critical aspect of the preprocessing phase involved identifying and addressing missing values within the dataset. Nine variables exhibited missing values, with "weight" showing the highest proportion at 96.9%, followed by "max_glu_serum" and "A1Cresult" at 94.75% and 83.28% respectively. (Figure 2.2)

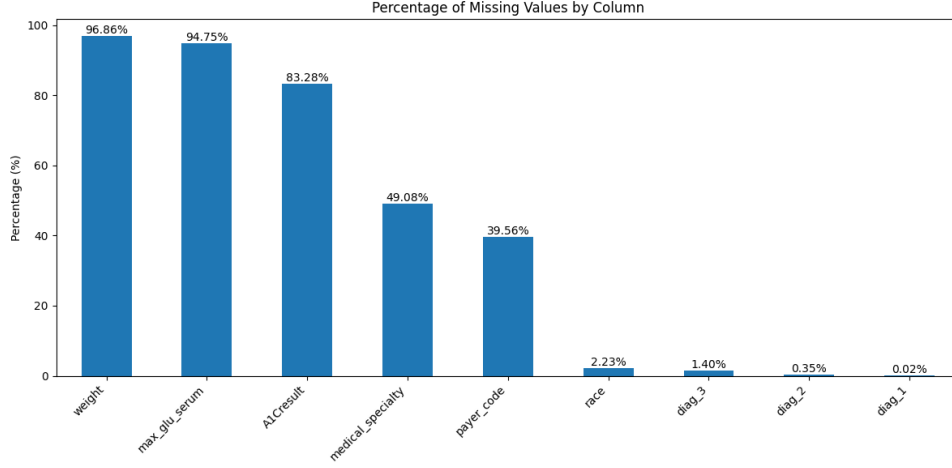


Figure 2.2: Percentage of missing values by variable

Although "weight" is very important factor to identify the diabete condition, however, due to its extremely high percentage of missing values (96.9%) we dropped this variable. Despite its high percentage of missing values, due to small amount of categorical objects, the "max_glu_serum", "A1Cresult", and "medical_specialty", were imputed with the value "None". Similar to the above categorical variables, the "payer_code" and "race" variables were imputed with the value "None". For diagnosis variables: "diag_1", "diag_2", "diag_3", missing values were imputed with the value "None". Categories with less than 0.5% occurrence were merged into a new "Rest" category to reduce dimensionality and prevent overfitting. Due to its importance, the "diag_1" variable was considered more important than "diag_2", and "diag_2" was considered more important than "diag_3". If a patient has multiple diagnoses, the most important one was selected and put into the "diag" variable. The "Rest" category was created to handle the rare cases and thus has less importance ranking (after diag_3). The "None" category was created to handle the missing values and has the least importance (after "Rest"). This approach was used to handle the overfitting issue and reduce the dimensionality of the dataset. For "race", the missing values were imputed with "None", although it could be aligned with "Caucasian" or "AfricanAmerican" categories, due to high median values. For "age", the missing values were imputed with the median of the variable. ("age_mid")

Based on distribution analysis and correlation with the target variable, we retain the following categorical variables: Demographics: race, gender, payer_code (Figure 2.3), Medical details: max_glu_serum, A1Cresult, medical_specialty (Figure 2.4), Medications: metformin, glimepiride, glipizide, glyburide, pioglitazone, rosiglitazone (Figure 2.5), Treatment process: insulin,change, diabetesMed (Figure 2.6), Diagnosis: diag (Figure 2.7), Hospital factors: admission_type_id, discharge_disposition_id, admission_source_id (Figure 2.8)

From above analysis, we can see that: Over 70% of patients are Caucasian, with distinct readmission patterns across racial groups (Figure 2.3), For Emergency intake (type 1) accounts for more than 50% of admissions (Figure 2.8) For Discharge disposition: Approximately 60% of patients are discharged to home (Figure 2.8) Lastly, For Admission source: Nearly 60% of admissions originate from emergency rooms, followed by 30% from physician referrals (Figure 2.8)

Detailed information about the codes of Admission type, Discharge disposition, and Admission source are shown in Table 2.1, Table 2.2, and Table 2.3.

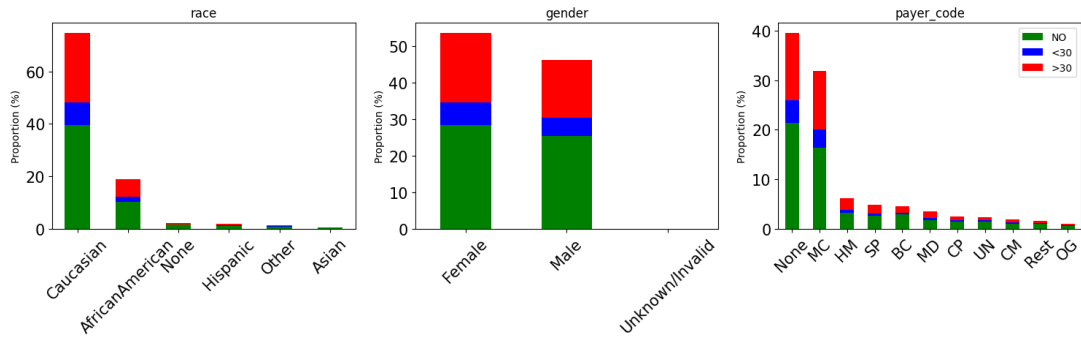


Figure 2.3: Distribution of readmission status across Demographics

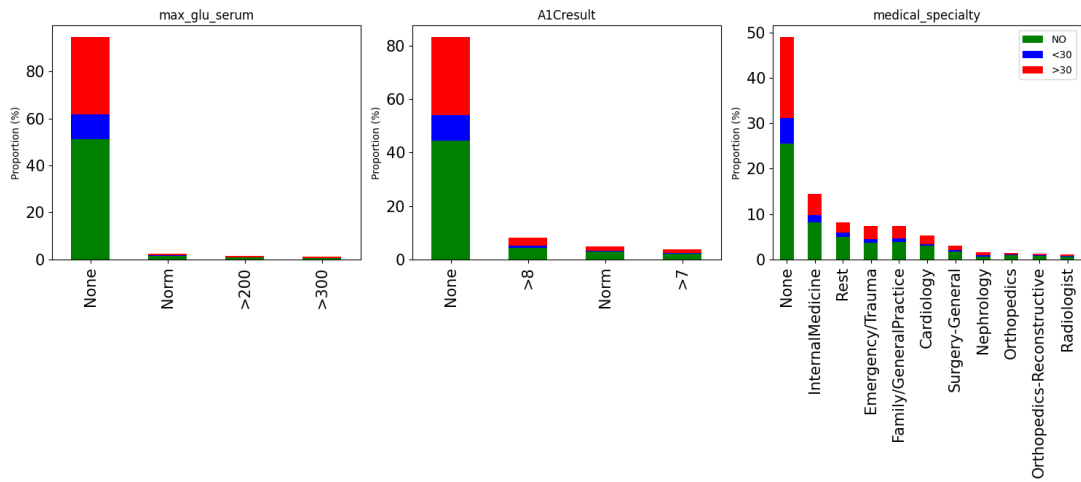


Figure 2.4: Distribution of readmission status across Medical details

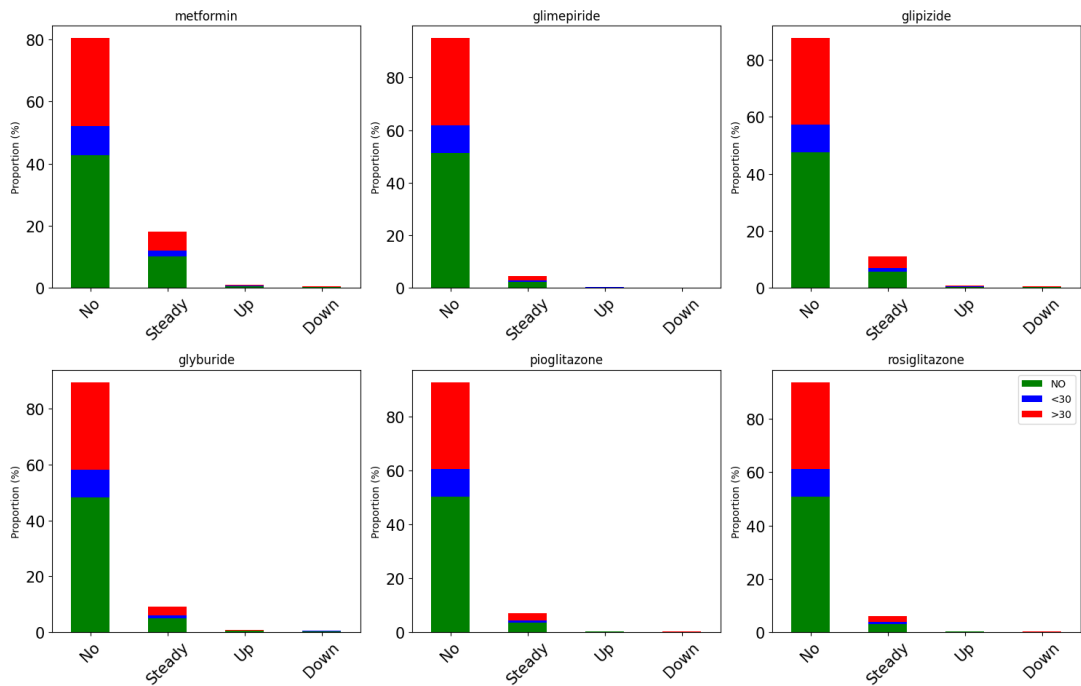


Figure 2.5: Distribution of readmission status across Medications

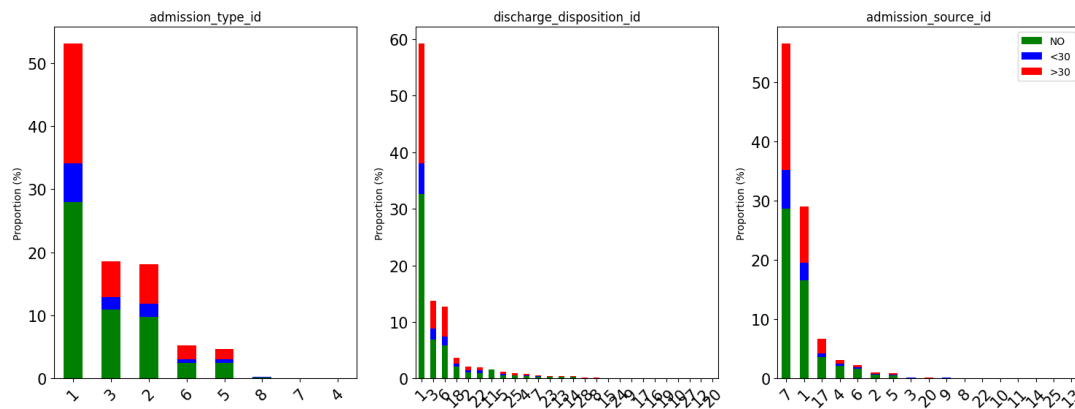
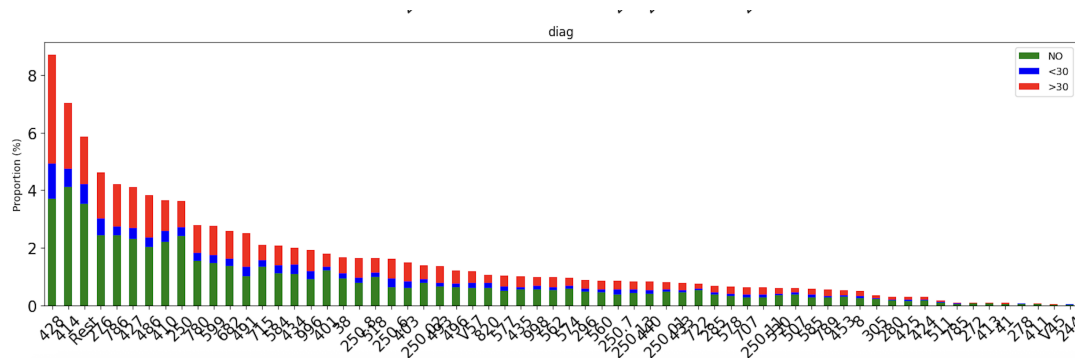
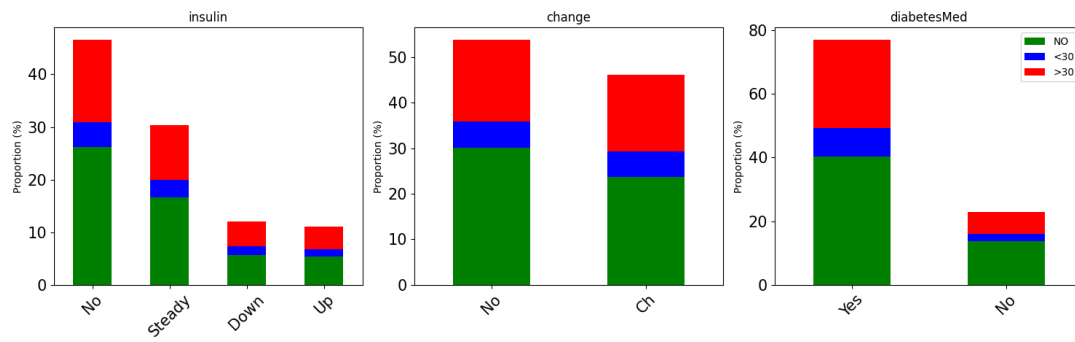


Table 2.1: Admission types and their descriptions

admission_type_id	description
1	Emergency
2	Urgent
3	Elective
4	Newborn
5	Not Available
6	NULL
7	Trauma Center
8	Not Mapped

Table 2.2: Discharge Disposition Descriptions

discharge_disposition_id	description
1	Discharged to home
2	Discharged/transferred to another short term hospital
3	Discharged/transferred to SNF
4	Discharged/transferred to ICF
5	Discharged/transferred to another type of inpatient care institution
6	Discharged/transferred to home with home health service
7	Left AMA
8	Discharged/transferred to home under care of Home IV provider
9	Admitted as an inpatient to this hospital
10	Neonate discharged to another hospital for neonatal aftercare
11	Expired
12	Still patient or expected to return for outpatient services
13	Hospice / home
14	Hospice / medical facility
15	Discharged/transferred within this institution to Medicare approved swing bed
16	Discharged/transferred/referred another institution for outpatient services
17	Discharged/transferred/referred to this institution for outpatient services
18	NULL
19	Expired at home. Medicaid only, hospice.
20	Expired in a medical facility. Medicaid only, hospice.
21	Expired, place unknown. Medicaid only, hospice.
22	Discharged/transferred to another rehab fac including rehab units of a hospital.
23	Discharged/transferred to a long term care hospital.
24	Discharged/transferred to a nursing facility certified under Medicaid but not certified under Medicare.
25	Not Mapped
26	Unknown/Invalid
27	Discharged/transferred to a federal health care facility.
28	Discharged/transferred/referred to a psychiatric hospital of psychiatric distinct part unit of a hospital
29	Discharged/transferred to a Critical Access Hospital (CAH). 7
30	Discharged/transferred to another Type of Health Care Institution not Defined Elsewhere

Table 2.3: Admission Source Descriptions

admission_source_id	description
1	Physician Referral
2	Clinic Referral
3	HMO Referral
4	Transfer from a hospital
5	Transfer from a Skilled Nursing Facility (SNF)
6	Transfer from another health care facility
7	Emergency Room
8	Court/Law Enforcement
9	Not Available
10	Transfer from critical access hospital
11	Normal Delivery
12	Premature Delivery
13	Sick Baby
14	Extramural Birth
15	Not Available
17	NULL
18	Transfer From Another Home Health Agency
19	Readmission to Same Home Health Agency
20	Not Mapped
21	Unknown/Invalid
22	Transfer from hospital inpt/same fac reslt in a sep claim
23	Born inside this hospital
24	Born outside this hospital
25	Transfer from Ambulatory Surgery Center
26	Transfer from Hospice

Several medication variables were excluded from further analysis due to extreme skewness that could lead to overfitting and they are represented in Figure [2.9](#)

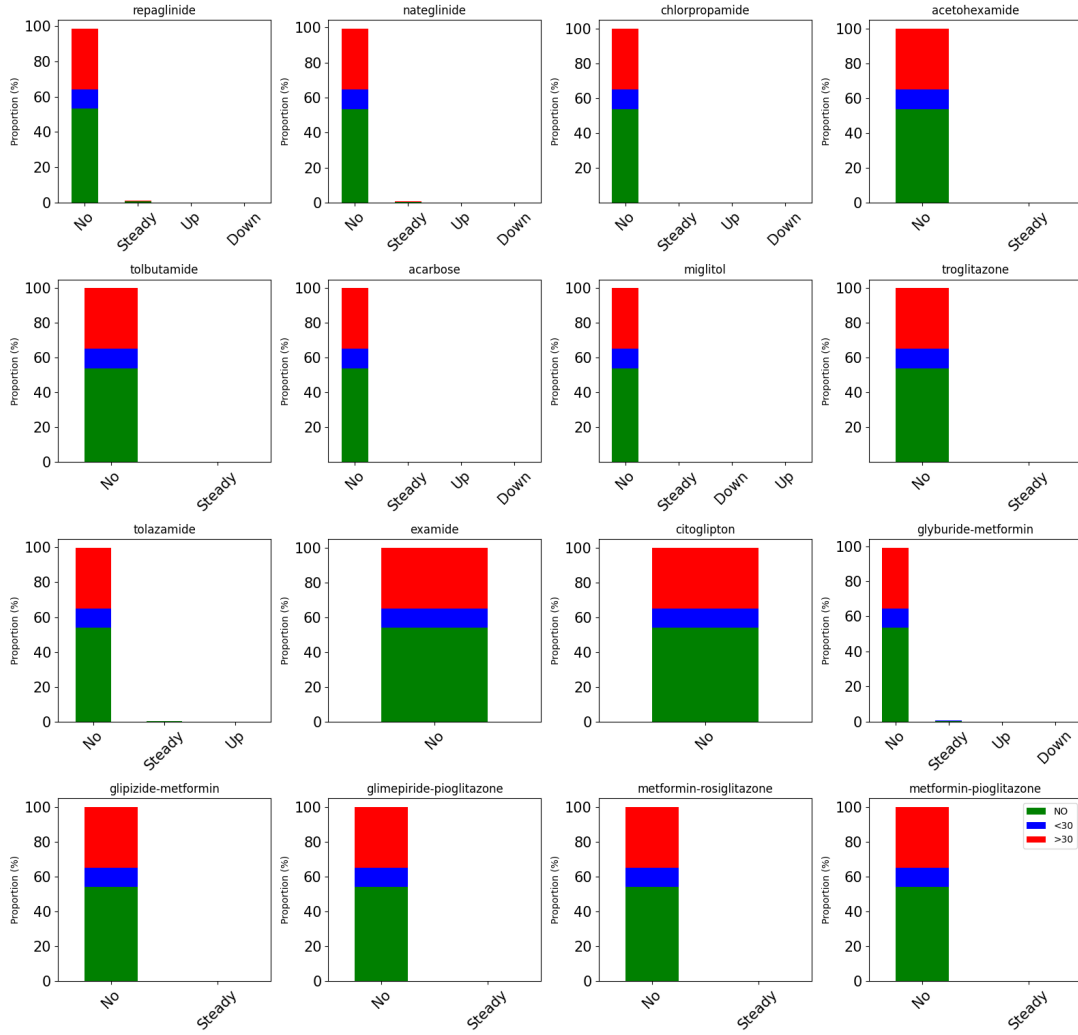


Figure 2.9: Distribution of readmission status across Other process

Next, we analyze the numerical variables in the dataset include: time_in_hospital, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, age_mid. For numerical data, cross correlation analysis was conducted to identify potential relationships between numerical variables. The correlation matrix did not reveal high collinearity between the variables, with most correlation coefficients below 0.4. (Figure 2.10)

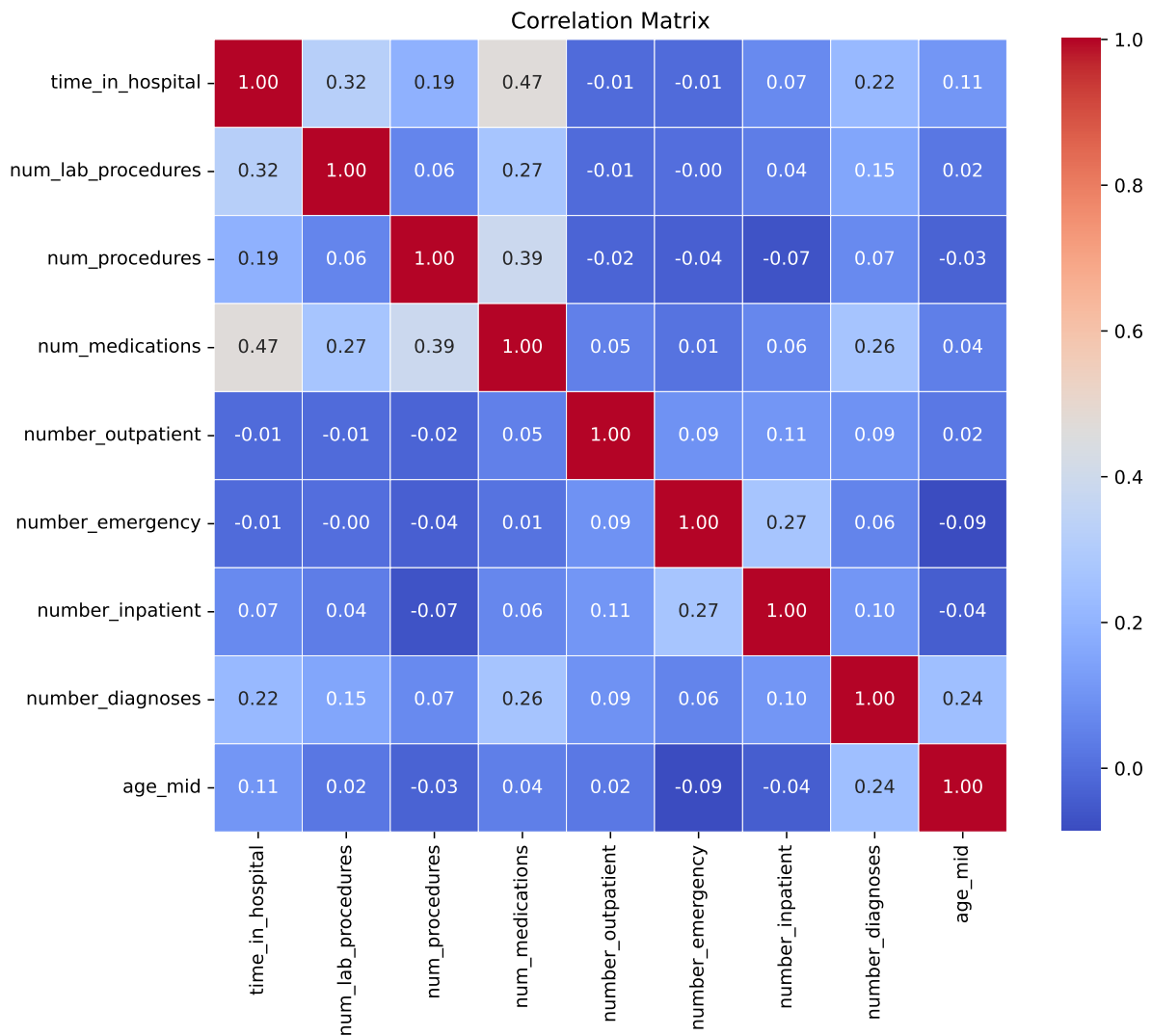


Figure 2.10: Correlation matrix of numerical variables

Variance Inflation Factor (VIF) analysis was also performed to detect multicollinearity. Although some potential collinearity was observed between "number_diagnoses" and "age", the VIF values remained within acceptable ranges (<10), and no numerical variables were removed based on this analysis. (Figure 2.11)

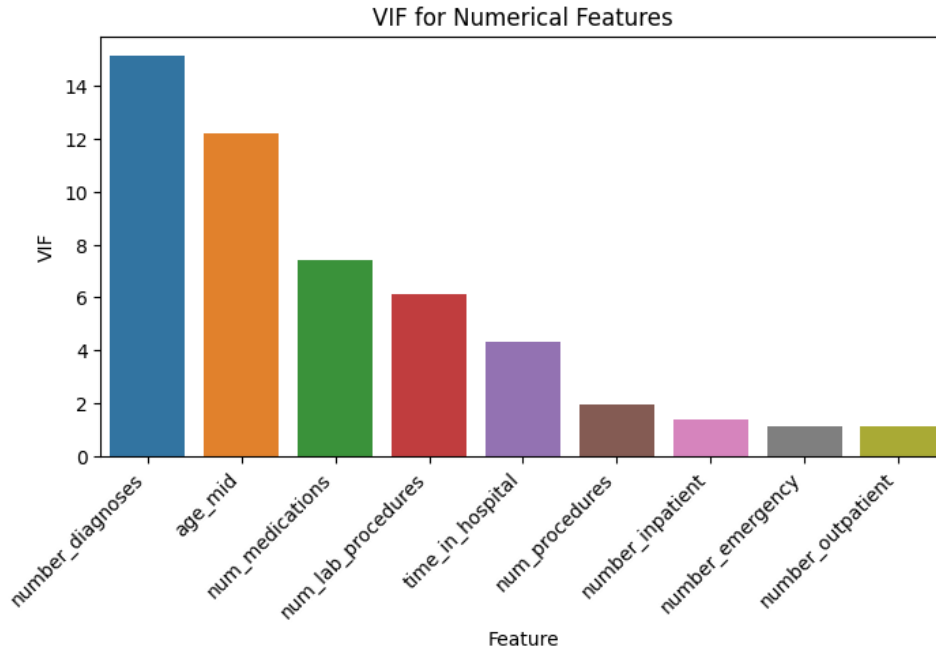


Figure 2.11: Variance Inflation Factor for numerical features

In addition to the correlation and VIF analysis, distribution analysis was conducted to identify potential probability distribution issues with the numerical variables. In Figure 2.12, histograms of numerical variables revealed the Gaussian distribution for num_lab_procedures; right-skewed distributions for time_in_hospital, num_procedures, and num_medications; left-skewed distribution for age_mid and number_diagnoses; and extreme right-skew for number_outpatient, number_emergency, and number_inpatient, with most values being zero.

Due to the extreme skew in outpatient, emergency, and inpatient visit counts, these variables were transformed into categorical variables: cat_outpatient: 0 for no visits, 1 for one or more visits; cat_emergency: 0 for no visits, 1 for one or more visits; cat_inpatient: 0 for no visits, 1 for one visit, 2 for more than one visit.

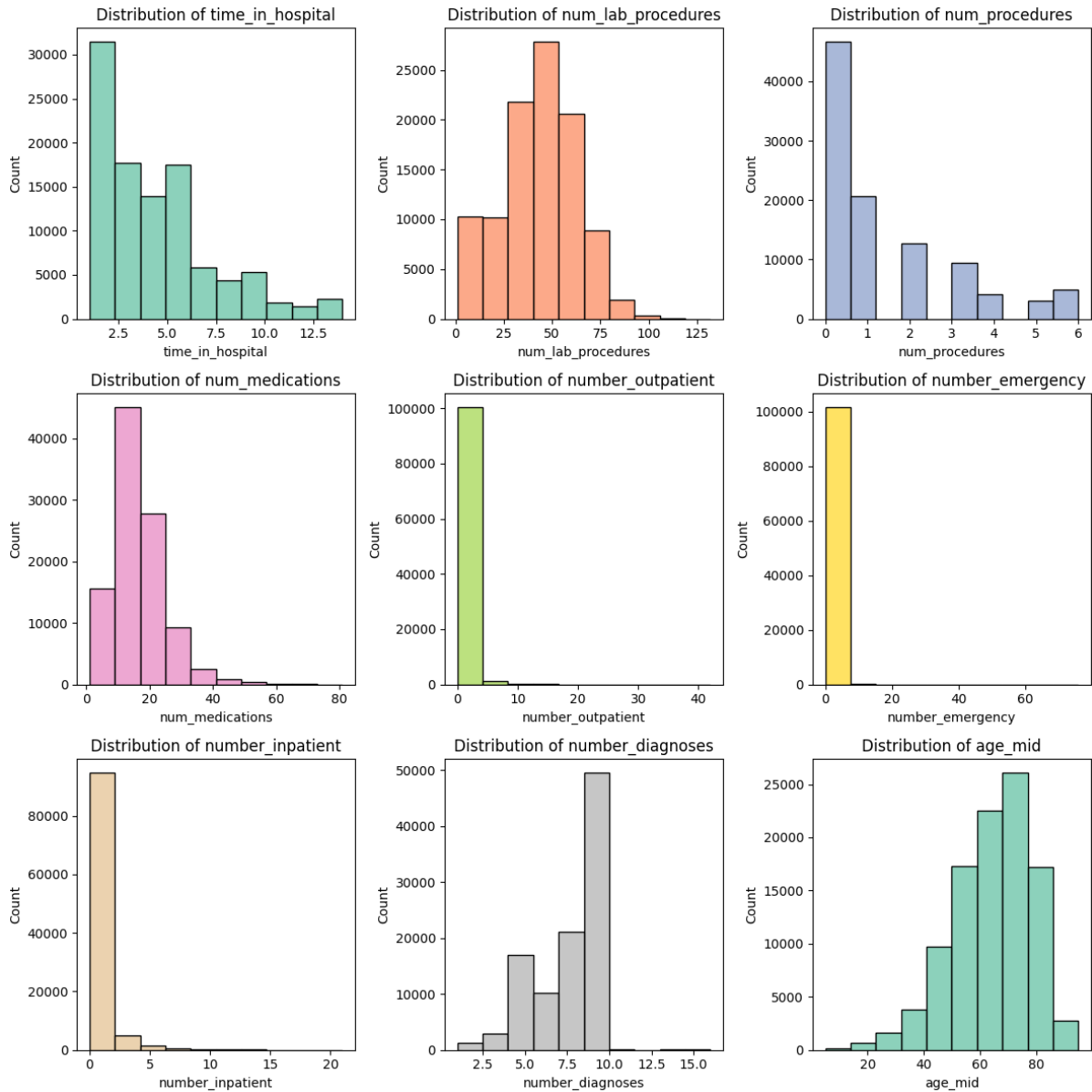


Figure 2.12: Distribution of numerical variables

The Boxplots in Figure 2.13 were used to visualize the relationship between numerical variables and the readmission status. The analysis showed comparable distributions across the three readmission categories, indicating that standardization of numerical variables would be appropriate before integration with categorical features.

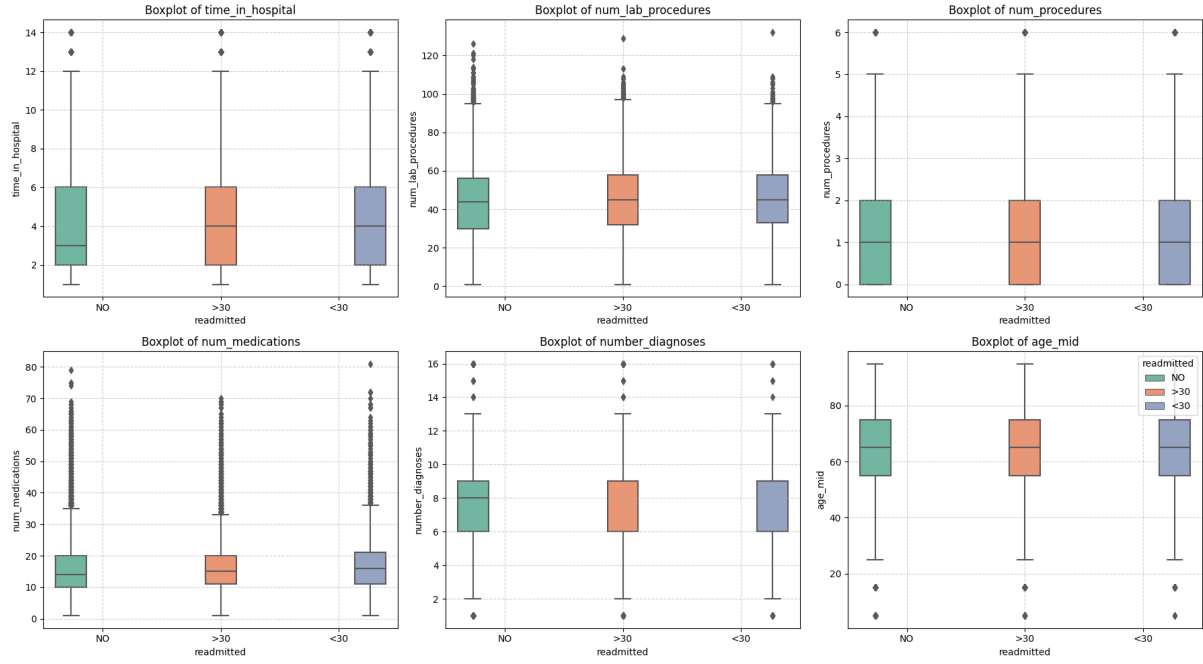


Figure 2.13: Boxplots of numerical variables by readmission status

After comprehensive exploratory analysis, we merged the transformed categorical variables (cat_outpatient, cat_emergency, cat_inpatient) with the selected categorical features. We applied One-hot encoding (OHE) to these categorical variables, resulting in 193 transformed features. We also scale the numerical variables then integrate to form the final input feature matrix X .

This preprocessing pipeline transformed the original dataset with 50 variables into a modeling-ready matrix while preserving the information content and addressing the identified data quality issues.

Chapter 3

Modeling and Results

Logistic regression was selected as the primary modeling approach for this analysis due to its interpretability, statistical robustness, and suitability for classification problems. As a parametric model, logistic regression provides clear insights into the relationship between predictors and the probability of hospital readmission, facilitating clinical interpretation and decision-making.

For this multiclass classification problem with three readmission categories ("NO", ">30", "<30"), we employed a multinomial logistic regression approach. This extension of the binary logistic model allows for the prediction of probabilities across multiple classes.

Prior to model training, the preprocessed dataset was partitioned into training and testing sets using a stratified random sampling approach to maintain the class distribution of the target variable. The data was split with 80% allocated to the training set and 20% to the testing set.

To ensure robust model evaluation and mitigate the risk of overfitting, we applied 10-Fold cross-validation for the model training with `LogisticRegressionCV` in `scikit-learn`. Cross-validation provides a more reliable estimate of model performance by evaluating it on multiple subsets of the data, reducing the impact of sampling variability on performance assessment and also to avoid overfitting. Figure 3.1 shows the optimal regularization strength for each class and this regularization strength was used for the final model.

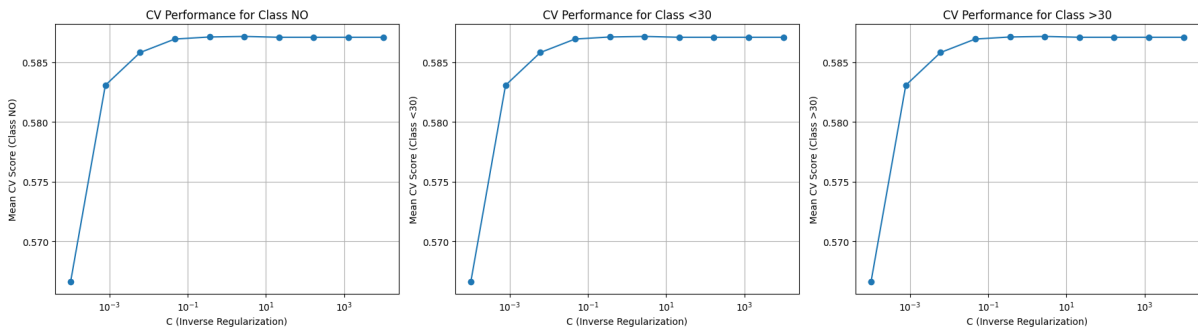


Figure 3.1: Confusion matrix for the logistic regression model on the test set

Once identified the optimal Regularization strength C from `LogisticRegressionCV`, the logistic regression model was trained using the `scikit-learn` implementation with `lbfgs` solver, 1000 maximum iterations, multinomial multi-class strategy and accuracy scoring with all available cpus.

The model was evaluated on both the cross-validation folds and the held-out test set using multiple performance metrics including: Accuracy, F1-score, Confusion Matrix, and Classification Report.

The model's classification report on the held-out test set in Table 3 and confusion matrix in Figure 3.2:

Class	Precision	Recall	F1-score
NO	0.61	0.84	0.71
>30	0.51	0.36	0.43
<30	0.23	0.01	0.02
Accuracy	0.58		

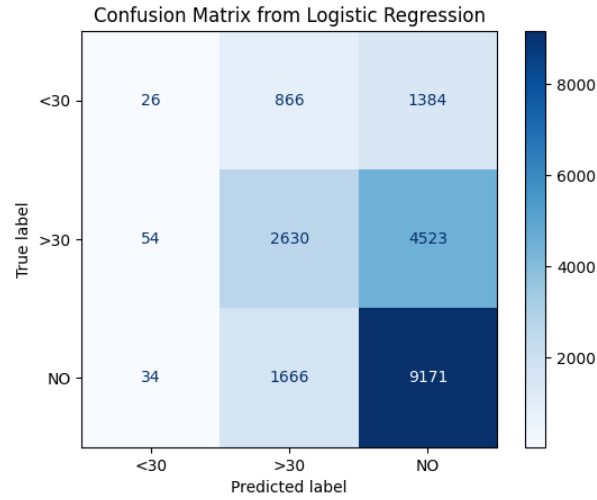


Figure 3.2: Confusion matrix for the logistic regression model

The model demonstrates stronger performance in identifying patients in "NO" class compared to those who will be readmitted. The more challenging class to predict is readmission within 30 days ("<30"), which has the lowest recall and F1-score. This is likely due to the class imbalance in the dataset, where the "<30" class represents only 11.2% of the observations.

The ROC curves on the other hand indicate better AUC score for all 3 classes is shown in Figure 3.3:

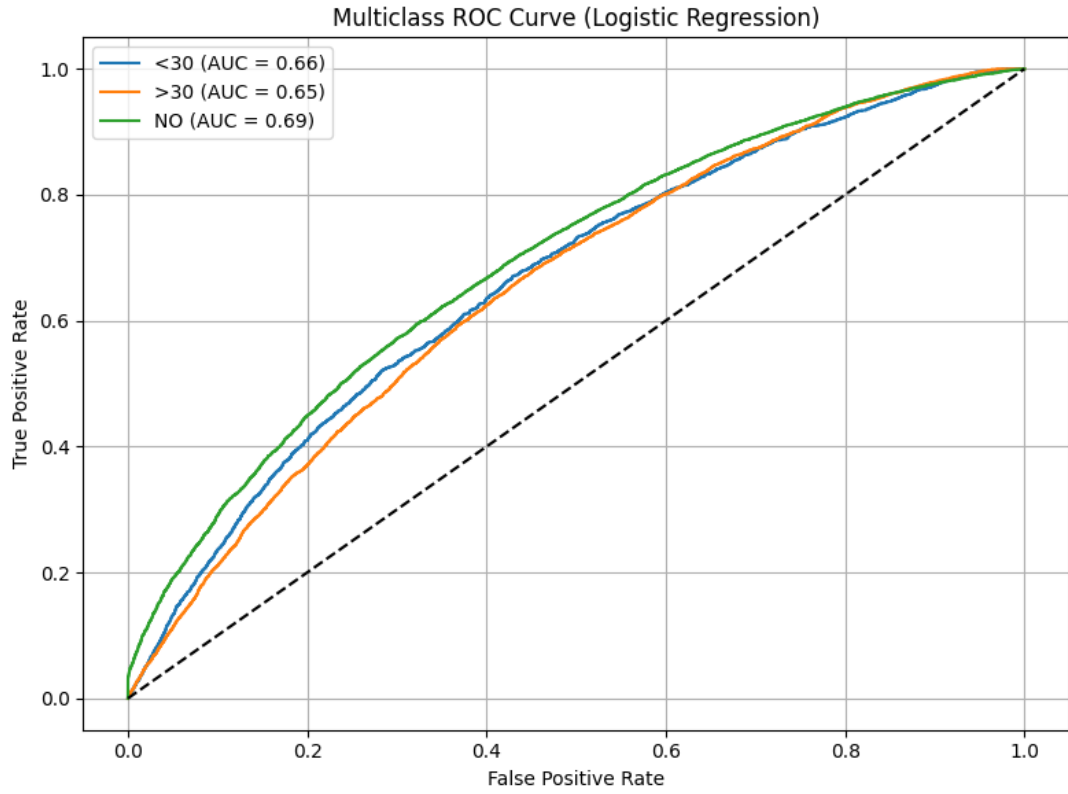


Figure 3.3: ROC curve for the logistic regression model on the test set

For multinomial logistic regression, the model produces a set of coefficients for each class. The magnitude and direction of these coefficients indicate the strength and nature of the relationship between each feature and the log-odds of belonging to a particular class (Figure 3.4).

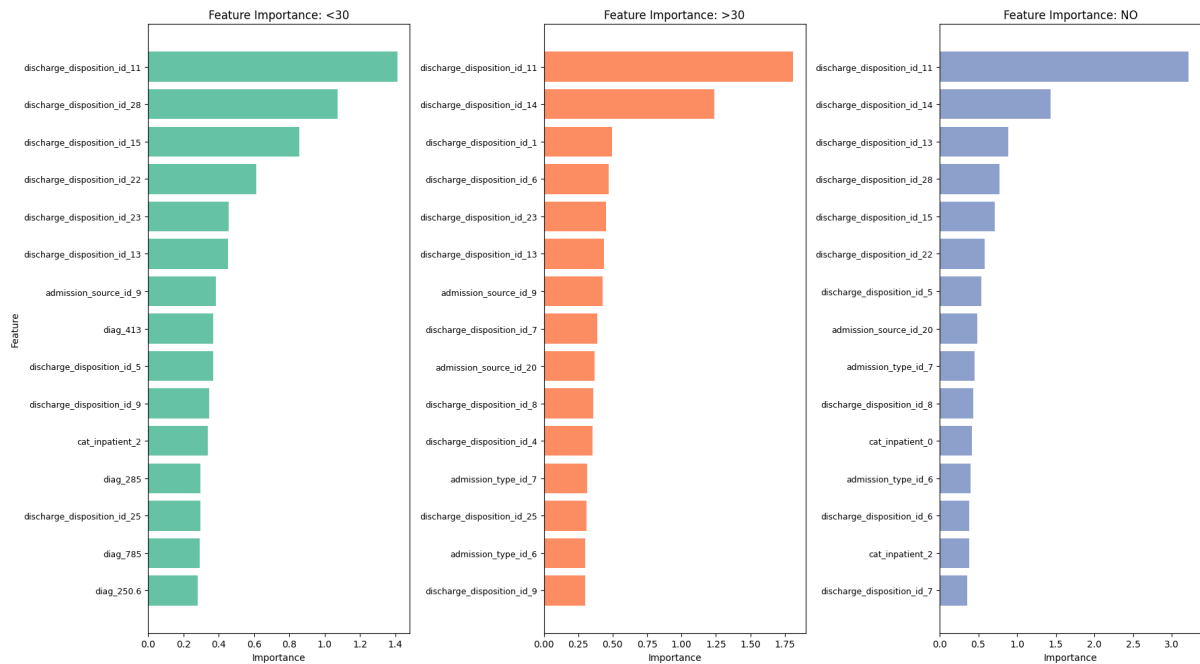


Figure 3.4: Top features by importance for each readmission category

The top 5 features for each class are among discharge disposition id code 11, 14, 15, 28 stands for "Expired", "Hospice", "Discharge within this institution", "Discharge to psychiatric facility" respectively. The Top features found from Logistic Regression are based on the coefficients of the model. The Logistic Regression model is a linear model and hence the coefficients are directly proportional to the importance of the features, assuming linear relationship between the features and the log-odds of the target variable. This hence makes the categorical features like discharge disposition stronger predictor.

The above feature analysis indicates that there exists potential non-linear relationship between the features and the target variable. Therefore, in order to find the right importance of the features, it is recommended to use the Random Forest approach (Figure 3.5): The importance features found from Random Forest model are more robust and can capture the non-linear relationship between the features and the target variable. Although the level of importance are not really strong but it is varying across predictors and provides more reliable results than the Logistic Regression model.

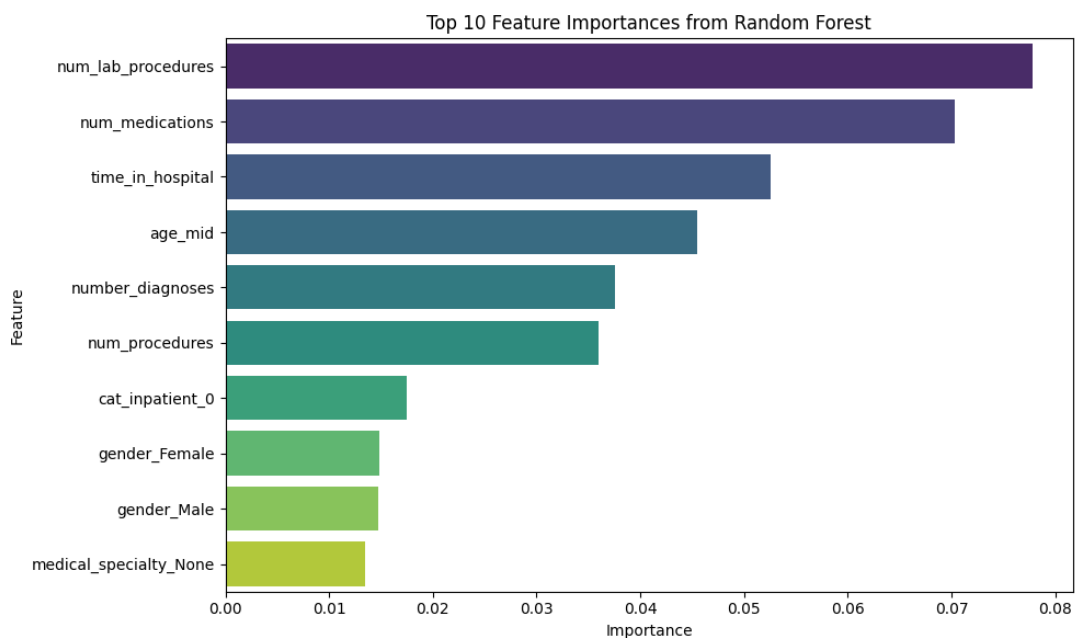


Figure 3.5: ROC curve for the logistic regression model on the test set using Random Forest

Chapter 4

Discussions and Conclusions

4.1 Discussions

This comprehensive analysis of diabetic patient data yielded several significant insights regarding hospital readmission patterns: The EDA revealed class data imbalance, with only 11.2% of patients readmitted within 30 days, compared to 34.9% readmitted after 30 days and 53.9% not readmitted. Missing value analysis identified "weight" as the most problematic variable with 96.9% missing data. Other variables with notable missingness but are included since they are categorical variables and the missing values can be imputed with None. Categorical variable analysis highlighted the significance of hospital administrative factors (admission type, discharge disposition, admission source) and medication regimens (particularly insulin, metformin, and glipizide) in differentiating readmission patterns. Numerical variable analysis demonstrated certain weak relationships between readmission and factors such as length of hospital stay, number of diagnoses, and previous hospital encounters, though none exhibited strong individual correlations with the target variable. The logistic regression model achieved moderate predictive performance with overall accuracy of 58% on the test set, demonstrating stronger capability in identifying non-readmissions than in predicting early readmissions. Feature importance analysis identified discharge disposition is the most significant feature in predicting the readmission (using Logistic Regression approach). On the other hand, the non-linear model Random Forest identified num_lab_procedures, num_medications, age, gender and time_in_hospital as key predictors of readmission.

4.2 Conclusion

This analysis of diabetic patient data has provided valuable insights into the complex interplay of factors associated with hospital readmission. The logistic regression model developed through this work offers a foundation to predict and reduce readmission rates. While the model demonstrates moderate predictive performance, its strength lies in the interpretability of its findings and their alignment with clinical understanding. The identified risk factors—particularly discharge disposition, previous hospitalizations, multiple diagnoses, and medication changes—represent actionable targets for quality improvement initiatives. By focusing on these aspects of care, healthcare providers can work toward reducing the substantial burden of hospital readmissions among diabetic patients, ultimately improving both clinical outcomes and healthcare resource utilization.

Further refinement of predictive models and prospective evaluation of targeted interventions based on these findings will be essential to translate this analytical work into meaningful improvements in diabetic patient care.